

5.1

EXOTHERMIC AND ENDOTHERMIC REACTIONS



CONTENTS

3 Stoichiometry

3.1 - Formulae

3.2 - Relative masses of atoms and molecules

3.3 - The mole and the Avogadro constant

4 Electrochemistry

4.1 - Electrolysis

4.2 - Hydrogen-oxygen fuel cells

5 Chemical energetics

5.1 - Exothermic and endothermic reactions

6 Chemical reactions

6.1 - Physical and chemical changes

6.2 - Rate of reaction

6.3 - Reversible reactions and equilibrium

6.4 - Redox

7 Acids, bases and salts

7.1 - The characteristic properties of acids and bases

7.2 - Oxides

7.3 - Preparation of salts

8 The Periodic Table

8.1 - Arrangement of elements

8.2 - Group I properties

8.3 - Group VII properties

5.1 Exothermic and endothermic reactions

Core

- 1 State that an exothermic reaction transfers thermal energy to the surroundings leading to an increase in the temperature of the surroundings
- 2 State that an endothermic reaction takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings
- 3 Interpret reaction pathway diagrams showing exothermic and endothermic reactions

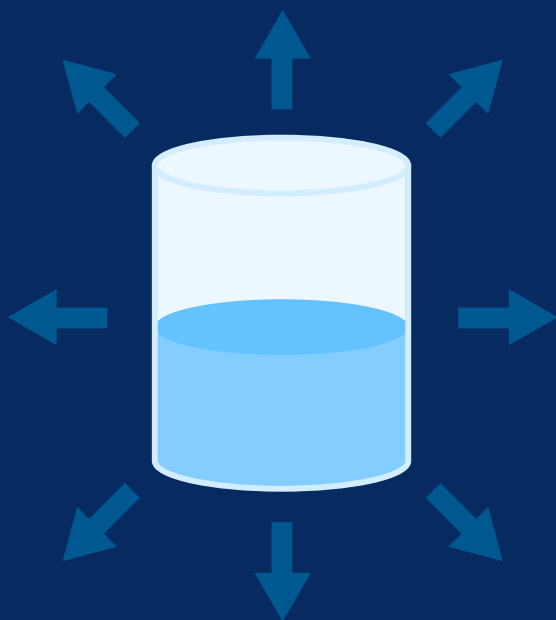
Supplement

- 4 State that the transfer of thermal energy during a reaction is called the enthalpy change, ΔH , of the reaction. ΔH is negative for exothermic reactions and positive for endothermic reactions
- 5 Define activation energy, E_a , as the minimum energy that colliding particles must have to react
- 6 Draw and label reaction pathway diagrams for exothermic and endothermic reactions using information provided, to include:
 - (a) reactants
 - (b) products
 - (c) enthalpy change of the reaction, ΔH
 - (d) activation energy, E_a
- 7 State that bond breaking is an endothermic process and bond making is an exothermic process and explain the enthalpy change of a reaction in terms of bond breaking and bond making
- 8 Calculate the enthalpy change of a reaction using bond energies

Exothermic and endothermic reactions

State that an **exothermic reaction** transfers thermal energy to the surroundings leading to an increase in the temperature of the surroundings

State that an **endothermic reaction** takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings



exothermic

‘exo’ means outside
‘thermic’ refers to heat



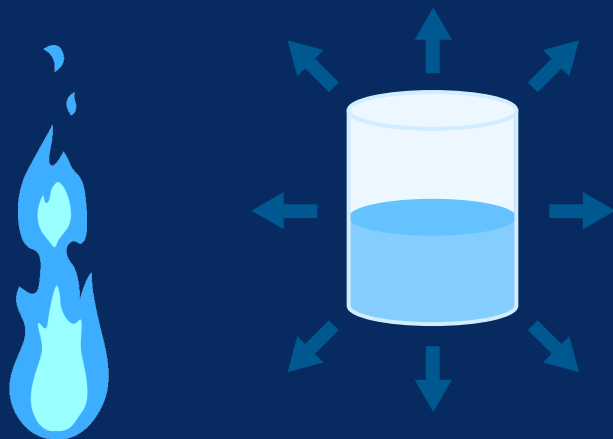
endothermic

‘endo’ means inside
‘thermic’ refers to heat

Exothermic and endothermic reactions

State that an **exothermic reaction** transfers thermal energy to the surroundings leading to an increase in the temperature of the surroundings

State that an **endothermic reaction** takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings



exothermic

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Exothermic and endothermic reactions

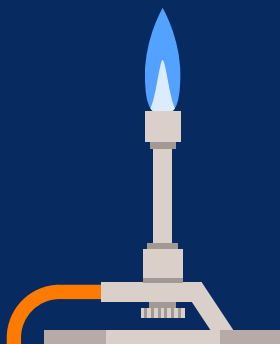
State that an exothermic reaction transfers thermal energy to the surroundings leading to an increase in the temperature of the surroundings

State that an **endothermic reaction** takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings



endothermic

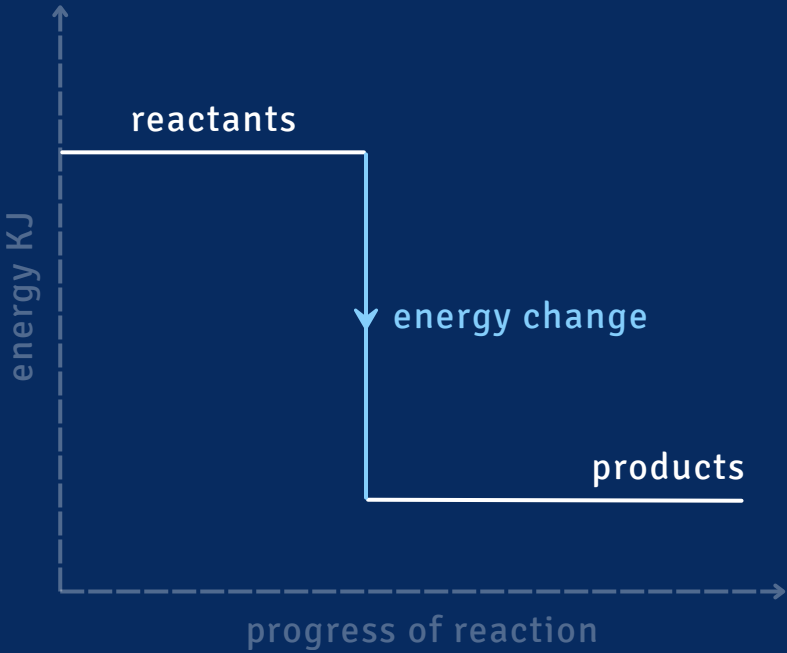
‘endo’ means inside
‘thermic’ refers to heat



Reaction pathway diagrams

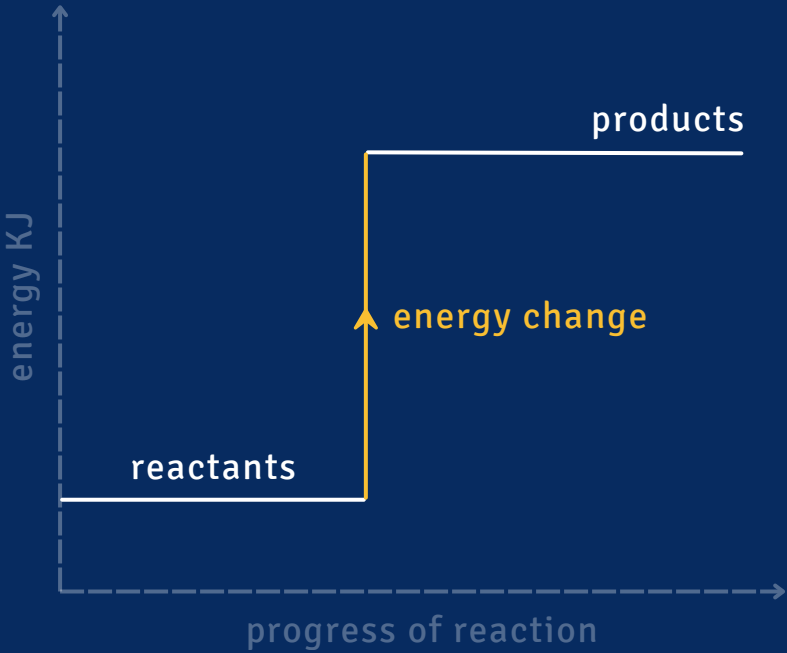
Interpret **reaction pathway diagrams** showing exothermic and endothermic reactions

(energy level diagrams)



exothermic

products have less energy than reactants



endothermic

products have more energy than reactants

Enthalpy change

EXTENDED

State that the transfer of thermal energy during a reaction is called the **enthalpy change**, ΔH , of the reaction

$$\Delta H = \begin{array}{c} \downarrow \downarrow \downarrow \\ \text{energy taken in} \end{array} - \begin{array}{c} \uparrow \uparrow \uparrow \\ \text{energy given out} \end{array}$$

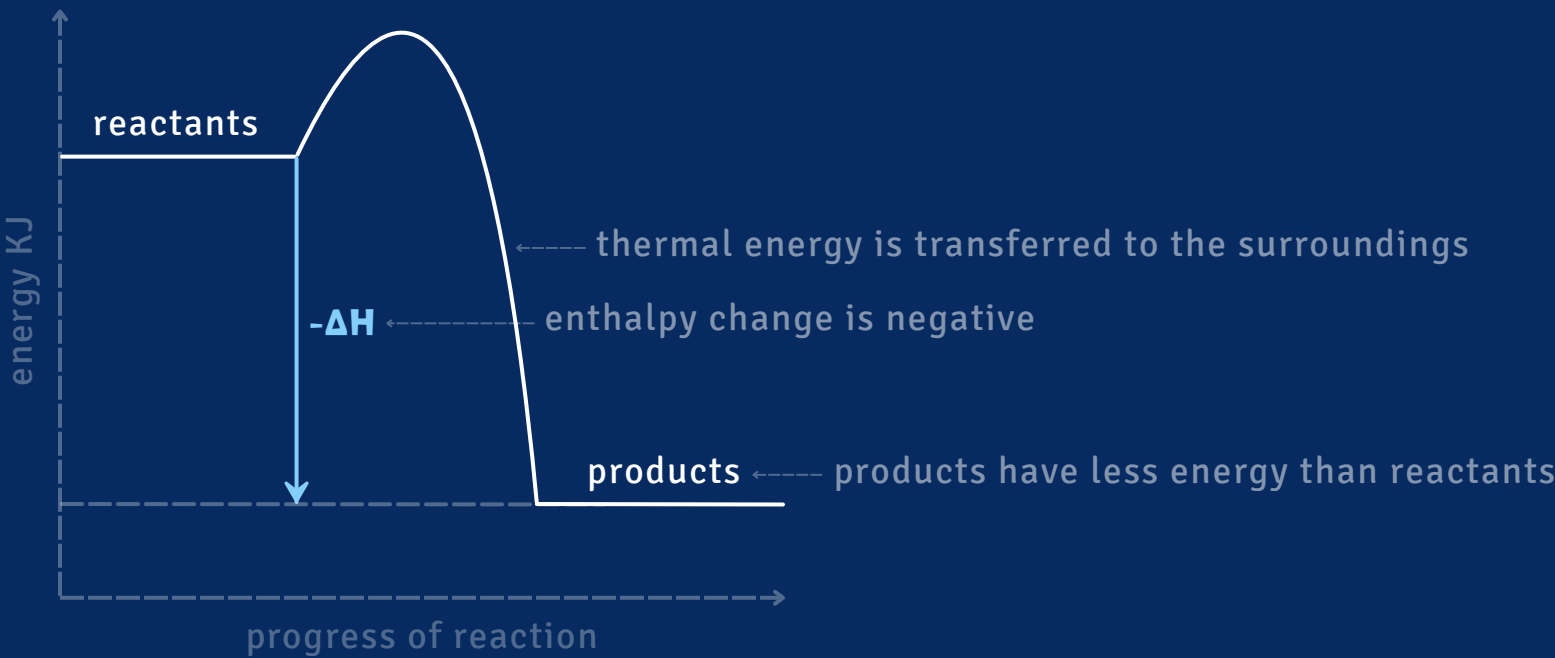
EXTENDED

Enthalpy change

State that the transfer of thermal energy during a reaction is called the **enthalpy change**, ΔH , of the reaction

- ΔH is negative for **exothermic** reactions

$$\Delta H = \text{energy taken in} - \text{energy given out}$$



EXTENDED

Enthalpy change

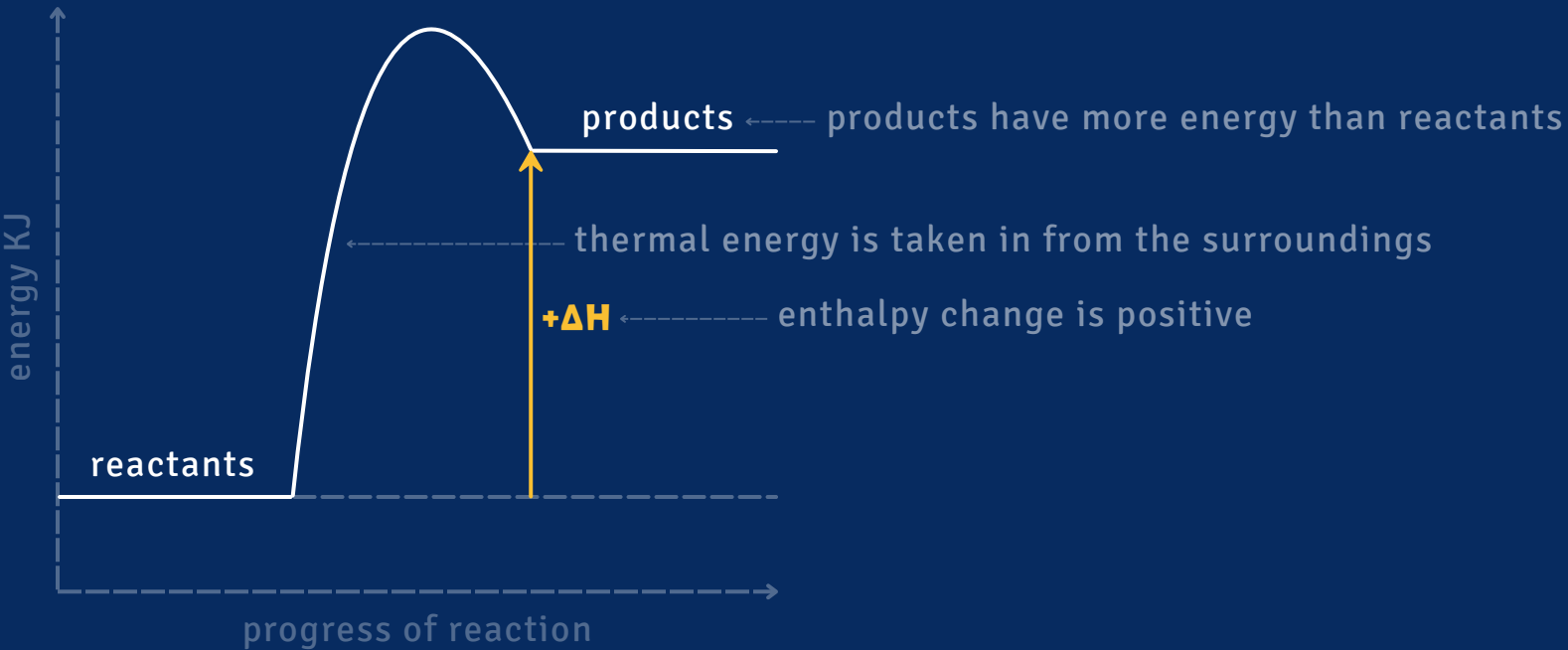
State that the transfer of thermal energy during a reaction is called the **enthalpy change**, ΔH , of the reaction

- ΔH is negative for **exothermic** reactions
- ΔH is positive for **endothermic** reactions





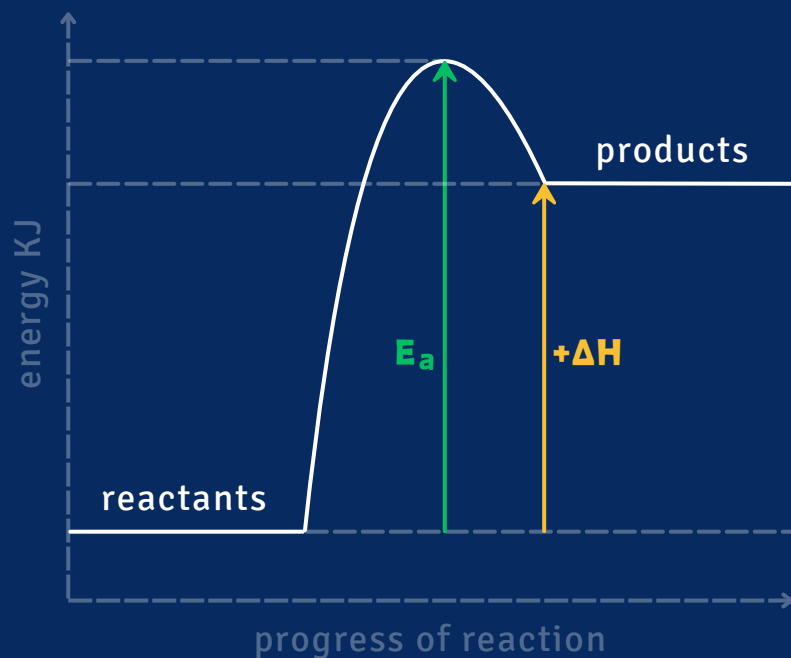
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Activation energy

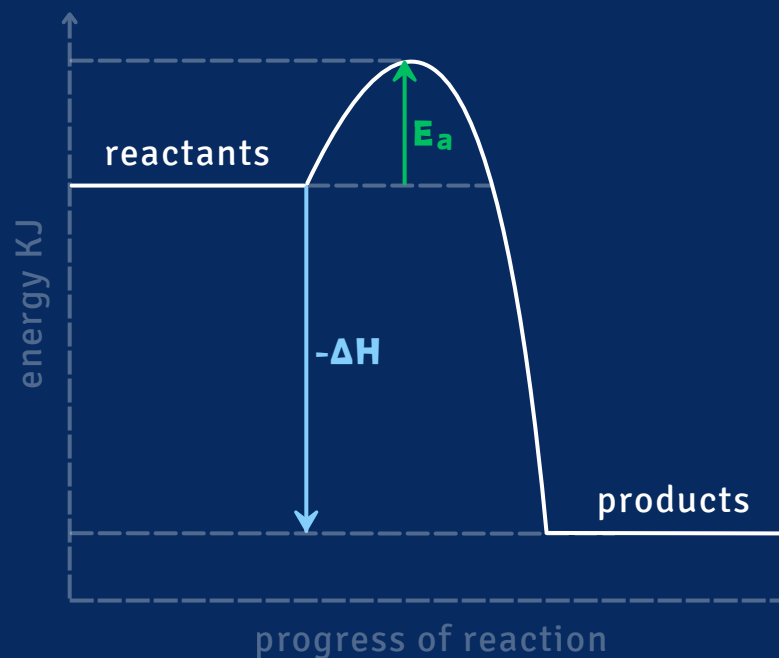
EXTENDED

Define **activation energy**, E_a , as the minimum energy that colliding particles must have to react



endothermic

larger activation energy



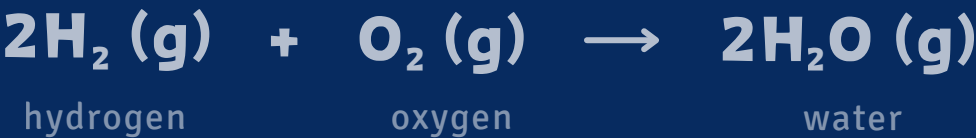
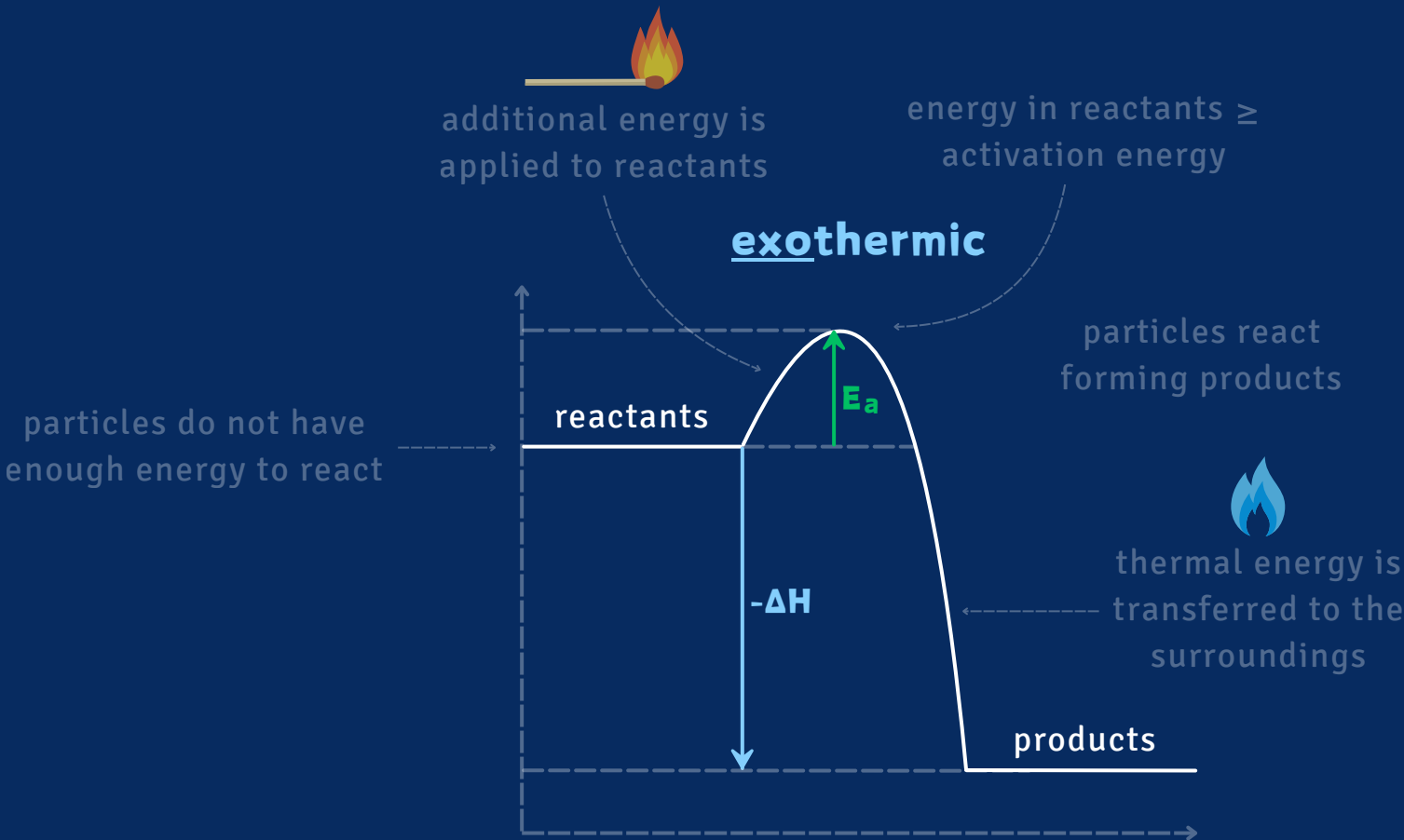
exothermic

smaller activation energy

EXTENDED

Activation energy

Define **activation energy**, E_a , as the minimum energy that colliding particles must have to react



Reaction pathway diagrams

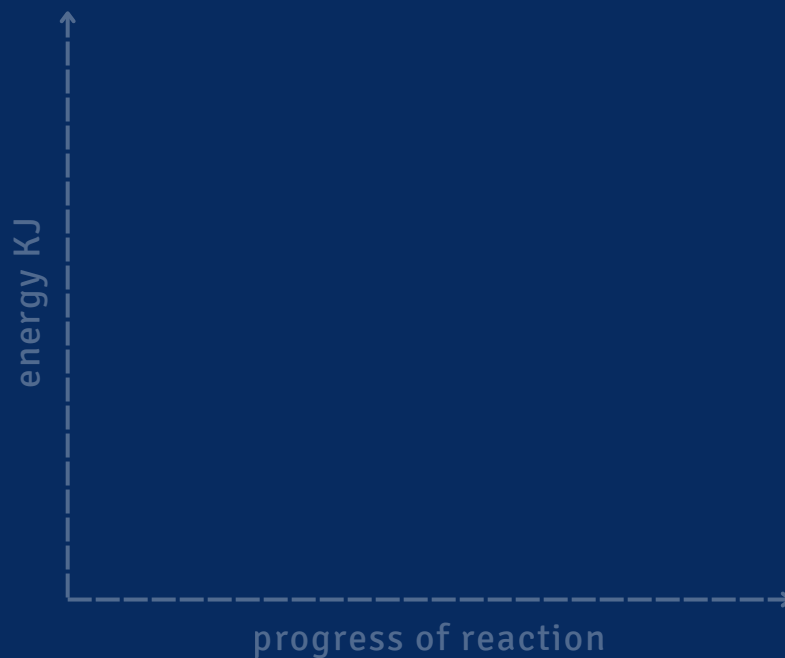
EXTENDED

Draw and label **reaction pathway diagrams** for endothermic and exothermic reactions, to include:

- reactants
- products
- enthalpy change of the reaction, ΔH
- activation energy, E_a



endothermic



exothermic

Reaction pathway diagrams

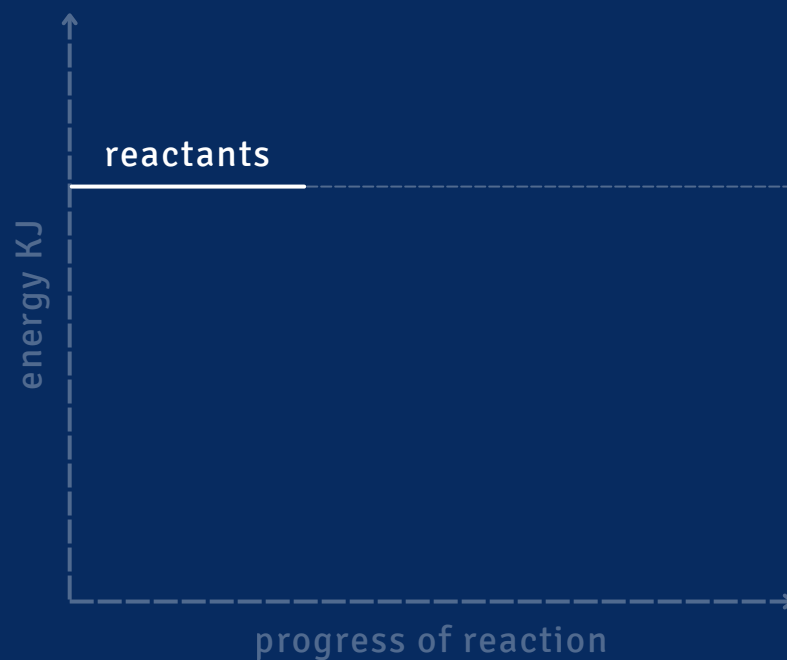
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endothermic



exothermic

EXTENDED

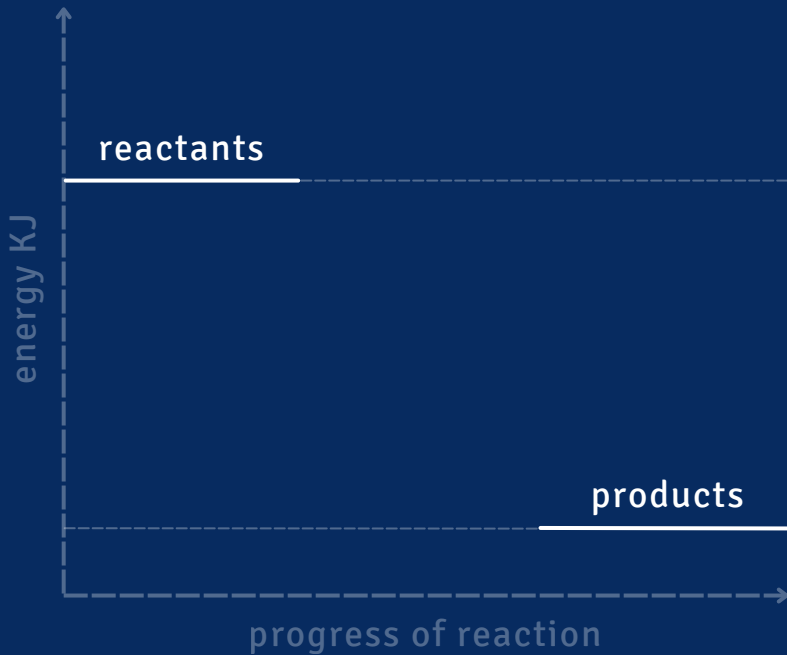
Reaction pathway diagrams

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endothermic



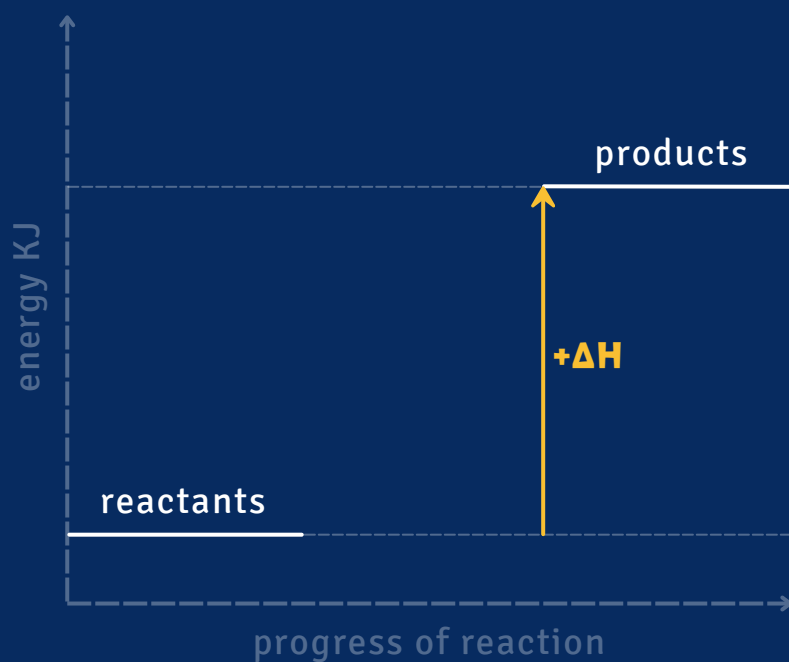
exothermic

Reaction pathway diagrams

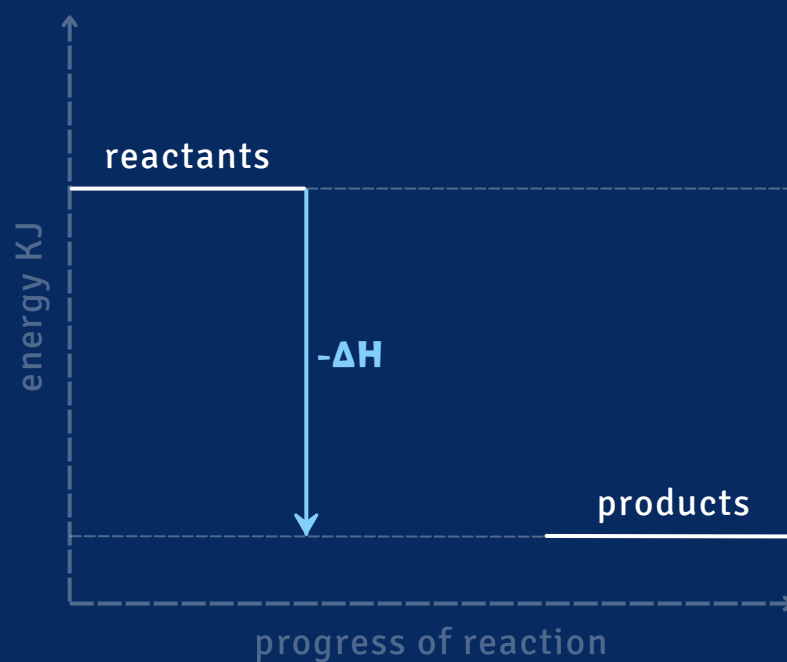
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endothermic



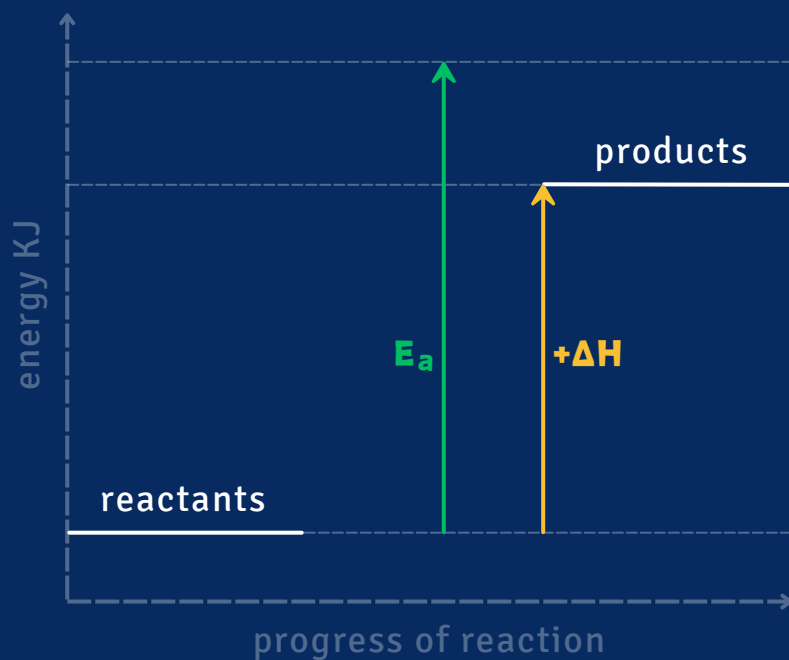
exothermic

Reaction pathway diagrams

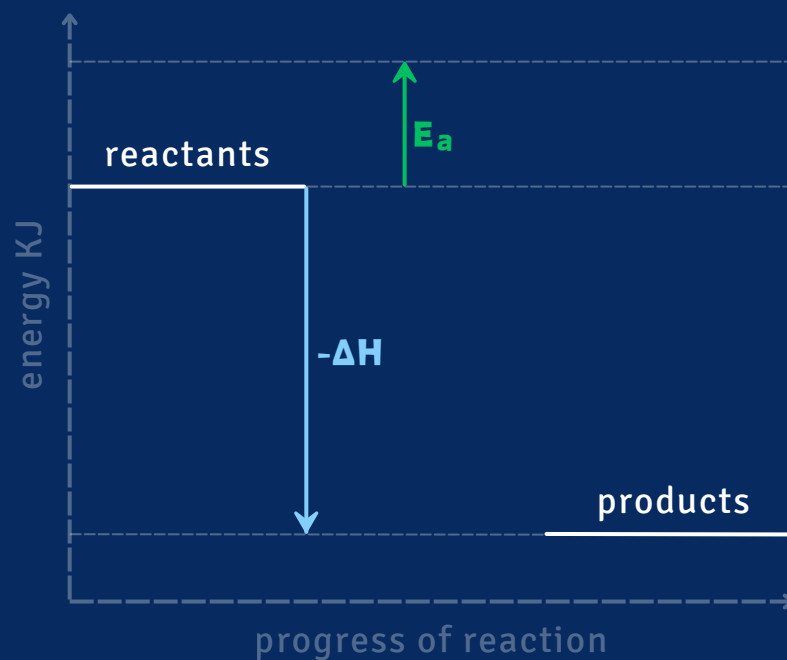
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endothermic



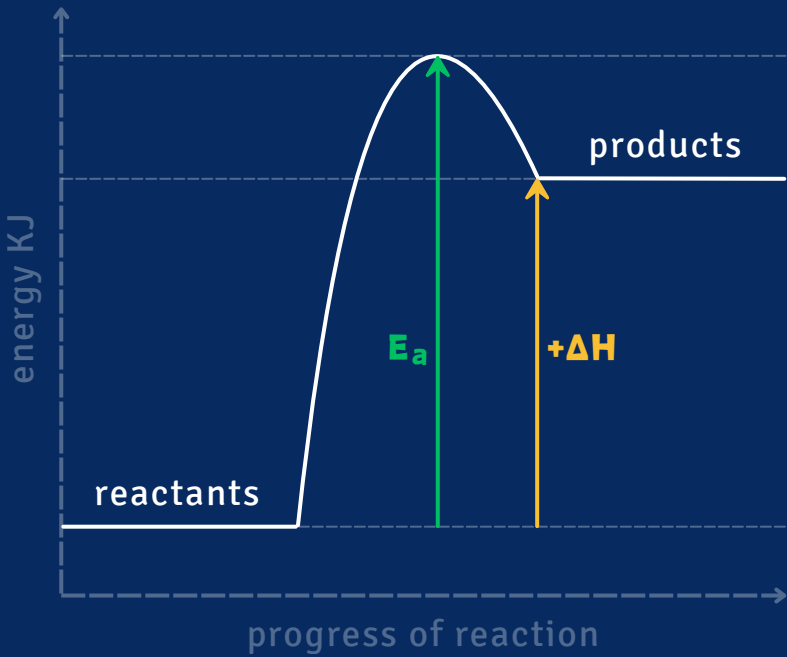
exothermic

EXTENDED

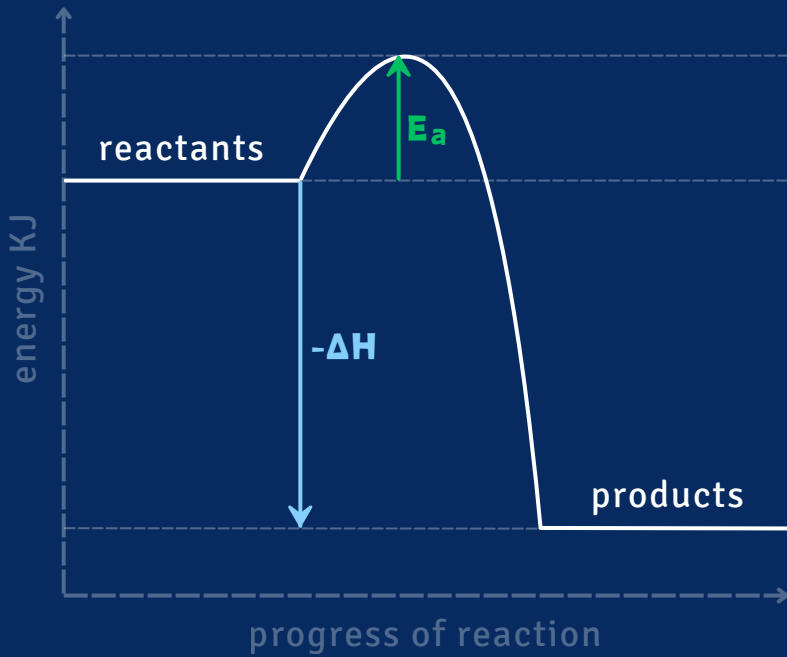
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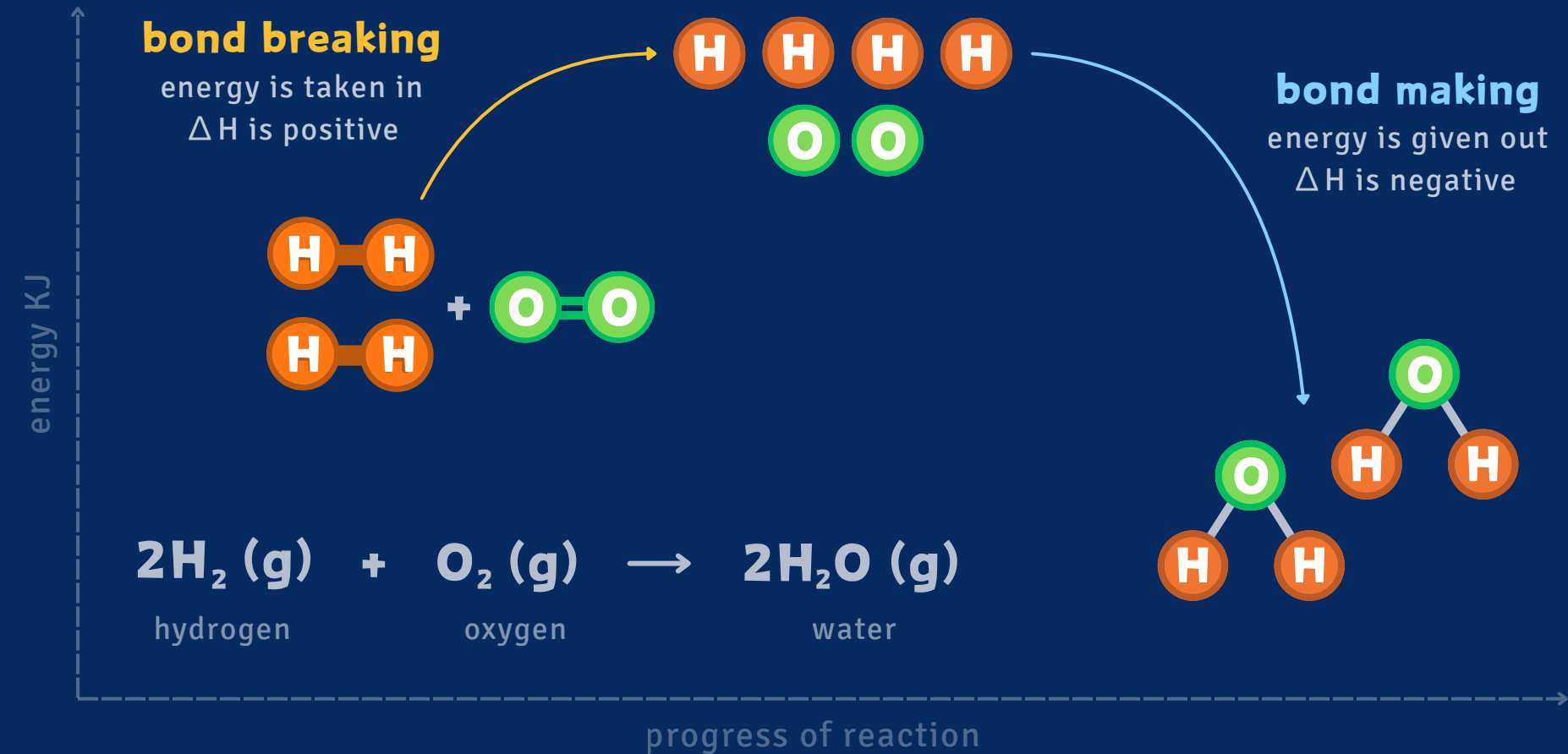


exothermic

EXTENDED

Bond breaking and bond making

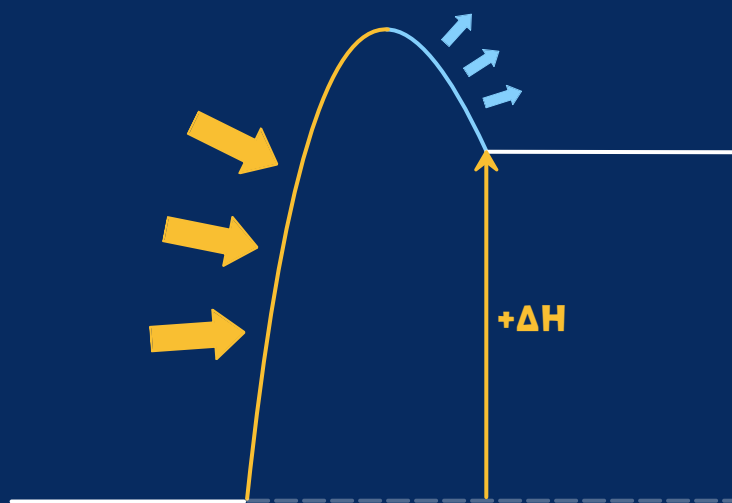
State that **bond breaking** is an endothermic process and **bond making** is an exothermic process



Bond breaking and bond making

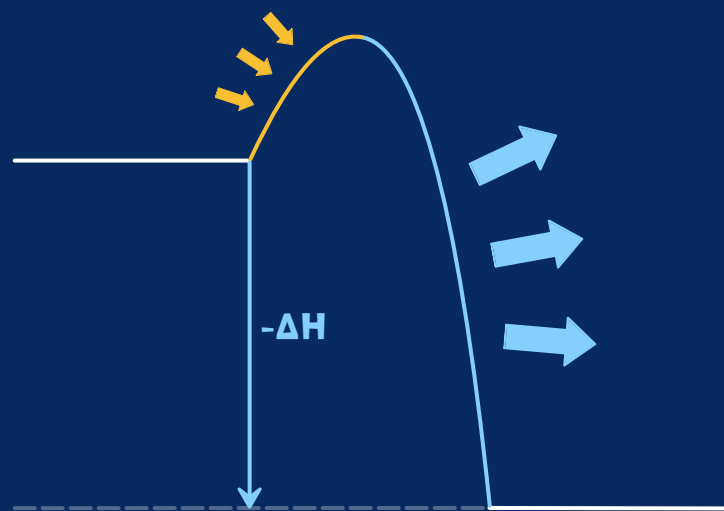
EXTENDED

Explain the **enthalpy change** of a reaction in terms of bond breaking and bond making



endothermic

energy in > energy out
 ΔH is positive



exothermic

energy in < energy out
 ΔH is negative

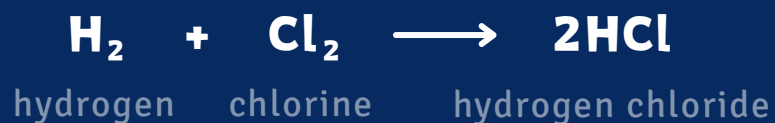
Enthalpy change calculations

EXTENDED

Calculate the **enthalpy change** of a reaction using **bond energies**

- bond energy** is the amount of energy required to break one mole of covalent bonds in gaseous molecules

Using the information provided, determine whether the reaction is exothermic or endothermic



bond	bond energy (KJ/mol)
H-H	435
Cl-Cl	242
H-Cl	432

$$\Delta H = \text{energy taken in} - \text{energy given out}$$

to break bonds
in reactants

when new bonds
form in products

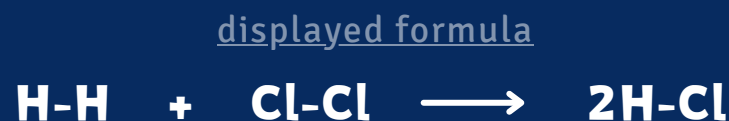
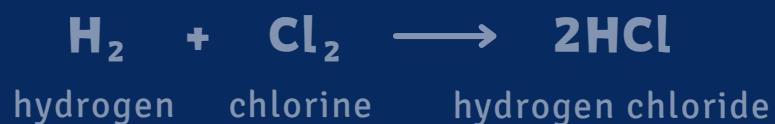
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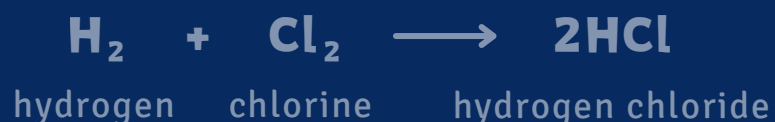
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Using the information provided, determine whether the reaction is exothermic or endothermic



displayed formula



bond	bond energy (KJ/mol)
H-H	435
Cl-Cl	242
H-Cl	432

$$\text{energy taken in} = 435 + 242 = \underline{677 \text{ KJ}}$$

$$\Delta H = \text{energy taken in} - \text{energy given out}$$

to break bonds
in reactants

when new bonds
form in products

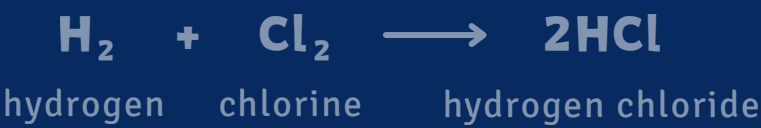
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bond	bond energy (KJ/mol)
H-H	435
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energy taken in = 435 + 242 = 677 KJ

energy given out = 2 x 432 = 864 KJ

$$\Delta H = \text{energy taken in} - \text{energy given out}$$

to break bonds when new bonds
in reactants form in products

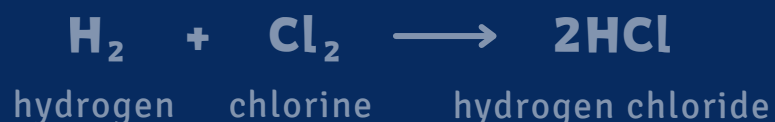
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displayed formula



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H-H	435
Cl-Cl	242
H-Cl	432

$$\text{energy taken in} = 435 + 242 = \underline{677 \text{ KJ}}$$

$$\text{energy given out} = 2 \times 432 = \underline{864 \text{ KJ}}$$

$$\Delta H = \text{energy taken in} - \text{energy given out}$$

$$\Delta H = 677 - 864$$

$$\Delta H = -187 \text{ KJ/mol} \quad \text{—} \quad \Delta H \text{ is negative, so the reaction is } \underline{\text{exothermic}}$$

5.1 Exothermic and endothermic reactions

Core

- 1 State that an exothermic reaction transfers thermal energy to the surroundings leading to an increase in the temperature of the surroundings
- 2 State that an endothermic reaction takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings
- 3 Interpret reaction pathway diagrams showing exothermic and endothermic reactions

Supplement

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